

A PROJECT REPORT ON

MEDGUARD AI

SUBMITTED TO
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BACHELOR OF TECHNOLOGY (Computer Science & Engineering/ Information Technology)

BY

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2025-26



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Hereby declare that the project work incorporated in the present project entitled **“MEDGUARD AI”** is original work. This work has not been submitted to any University for the award or a Degree or a Diploma. We have properly acknowledged the material collected from secondary sources wherever required. We solely own the responsibility for the originality of the entire content.

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EXAMINER'S APPROVAL CERTIFICATE

The project report entitled “MEDGUARD AI” submitted by Sudip Konde (MITU22BTCS0859), Tanmay Khedekar (MITU22BTCS0609), Sai Jagdale (MITU22BTCS0692), Sujal Bafna (MITU22BTCS0864) in partial fulfillment for the award of the degree of Bachelor of Technology (Computer Science & Engineering) during the academic year 2025-26, of MIT-ADT University, MIT School OF COMPUTING, Pune, is hereby approved.

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ABSTRACT

The spread of counterfeit and substandard drugs in India poses a serious threat to public health, as existing verification methods are often slow, manual, and unreliable. This project presents **MedGuard AI**, an advanced AI-powered medicine verification system designed to provide fast, intelligent, and reliable real-time drug authentication. It enables users to verify medicines by entering the name, uploading images (via OCR), or scanning barcodes, ensuring accessibility and ease of use.

At the core of MedGuard AI is an enhanced AI-driven workflow integrated with **Retrieval-Augmented Generation (RAG)** for intelligent data retrieval and verification. The OCR module (powered by Google Cloud Vision) extracts text from images, while the Barcode module decodes QR/barcodes to fetch product identifiers. The Extraction module processes key details such as Expiry Date, Manufacturing Date, and Price, generating automatic expiry alerts.

Unlike the previous approach using fuzzy matching, the system now leverages **RAG with PostgreSQL and pgvector**, enabling efficient similarity search and context-aware retrieval for accurate medicine verification. A Summary module, powered by a local LLM (Ollama), provides concise and user-friendly insights into medicine safety and composition.

The system is built using **Django REST Framework** with **JWT-based authentication**, ensuring secure and scalable multi-user access. The frontend is developed using **React with NodeJS**, enhancing responsiveness and user experience.

MedGuard AI offers a scalable, real-time, and practical solution for medicine verification, empowering users to make safe healthcare decisions.

Keywords: RAG, pgvector, PostgreSQL, Counterfeit Medicine Detection, OCR, Barcode Verification, JWT Authentication, React, Django REST Framework, Automatic Expiry Warning

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Chapter 1

INTRODUCTION

The increasing prevalence of counterfeit, expired, and substandard medicines poses a serious threat to global health and pharmaceutical integrity. Particularly in developing nations like India, poor-quality drugs can lead to ineffective treatments, severe side effects, and even death. These fake or degraded medicines erode public trust in healthcare systems and endanger patient lives. Despite numerous regulatory measures, existing verification tools remain manual, time-consuming, and unreliable, creating an urgent need for a fast, automated, and trustworthy system for drug authentication.

Traditional methods such as visual inspection or barcode scanning are insufficient to detect sophisticated counterfeits, as packaging and barcodes can easily be replicated. Laboratory testing, while accurate, is expensive and inaccessible for most consumers and pharmacists. Thus, a technological intervention is required that not only authenticates the product but also simplifies technical medical data for everyday users.

To address these challenges, MedGuard AI has been developed as an intelligent **AI-powered web application based on Django REST Framework** for real-time point-of-care drug verification. It provides three input modes:

1. Text Search
2. Image Upload
3. Barcode/QR Code Scan

MedGuard AI transforms this process into a multi-agent, orchestrated AI workflow, ensuring high accuracy, reliability, and speed. The system operates through a set of cooperating AI agents that perform distinct yet interconnected tasks under the control of an Orchestrator.

- The OCR Agent (“The Eyes”) employs Google Cloud Vision API to extract text from packaging.
- The Barcode Agent (“The Scanner”) decodes barcodes/QR codes and cross-verifies product identifiers.
- The Extraction Agent (“The Analyst”) extracts MFG, EXP, and MRP using Regex and generates expiry warnings.

• **The Retrieval Agent (“The Librarian”) replaces the earlier fuzzy matching approach and now uses Retrieval-Augmented Generation (RAG) with PostgreSQL and pgvector for semantic similarity search and accurate medicine identification.**

- The Summary Agent (“The Writer”), powered by a local LLM (Ollama – phi3), generates structured summaries (8-point for image/barcode, 5-point for text).

Furthermore, MedGuard AI includes a robust **JWT-based authentication system**, supporting login, signup, and personalized search history. This ensures privacy, security, and scalability. All user data is securely managed within a **PostgreSQL database integrated with pgvector for efficient retrieval operations**.

The frontend of the system is developed using **React with NodeJS**, providing an interactive and responsive user interface. The overall system follows a **microservice-based architecture**, enabling modularity, scalability, and real-time processing capabilities.

The major contribution in this article is:

- **Agentic AI Workflow:** A modular framework integrating OCR, Barcode, Extraction, Retrieval (RAG), and Summary agents.
- **Tri-Input Verification System:** Supports Text, Image, and Barcode-based verification.
- **Barcode Agent Integration:** Enables instant product validation using encoded identifiers.
- **RAG-Based Retrieval System:** Replaces fuzzy matching with pgvector-powered similarity search for improved accuracy and context-aware results.
- **Local LLM Summarization:** Generates structured summaries using Ollama (phi3).
- **Automated Safety Check:** Detects expiry and alerts users.
- **Multi-Layered Verification:** Combines OCR, Barcode, RAG retrieval, and LLM summarization.
- **Multi-User Platform:** Secure JWT authentication with personalized history.
- **Scalable Architecture:** Built with Django REST Framework, PostgreSQL, pgvector, and React.

Chapter 2

CONCEPT AND METHODS

The MedGuard AI system is built upon the foundation of Agentic Artificial Intelligence (AI) and Retrieval-Augmented Generation (RAG) — a modern approach to constructing reliable, explainable, and multi-step AI workflows. Unlike a single, monolithic model, the Agentic AI Workflow decomposes the problem into a sequence of specialized tasks, each performed by an independent “agent.” These agents operate in a sequential pipeline, where the output of one agent becomes the input for the next, ensuring robustness, clarity, and maintainability.

The system is orchestrated using Django, which serves as the backbone of user interaction and inter-agent communication. The overall workflow is divided into four core stages, each representing a distinct agent group with specific goals and computational methods.

2.1 Intake Agent (The Orchestrator)

Concept : The Intake Agent, implemented in Django’s views.py, functions as the central orchestrator that manages user interactions, determines workflow routes, and triggers the corresponding agents. It is responsible for handling all incoming requests and distributing them to the right processing modules (OCR, Barcode, or Text).

It acts as the system’s control centre, ensuring that the right agentic sequence is initiated based on the type of input the user provides.

Methods Used

- **Django Web Framework:** The project is structured as a Django web application, which manages user authentication (signup, login), URL routing (urls.py), and HTTP request handling. This allows a modular, multi-user environment where each user’s interactions are securely logged.
- **Multi-Modal Input Handling:** The home.html interface supports three input methods:
 1. **Text Input:** A search bar accepting direct medicine names.

2. Image Upload: A file upload system (multipart/form-data) for medicine package photos.
 3. Barcode/QR Code Scan: A camera or image-based input that decodes printed barcodes to fetch medicine identifiers instantly.
- Workflow Triage Logic: The orchestrator (views.py) inspects each incoming request.
 - If a barcode is detected → triggers the Barcode Agent workflow.
 - If an image file is detected → initiates the OCR + Extraction pipeline.
 - If plain text input is received → executes the Search + Summary workflow.

This design enables dynamic routing, ensuring each input type gets processed efficiently by the appropriate agents.

2.2 Vision & Extraction Agent (Data Digitization & Analysis)

Concept : This stage digitizes physical data from the medicine package and converts it into structured, machine-readable information. The process simulates human visual understanding using AI-driven OCR and analytical data extraction.

The Vision Agent captures text from the image, and the Extraction Agent processes it to retrieve essential fields — Manufacture Date (MFG), Expiry Date (EXP), and Maximum Retail Price (MRP).

Methods Used

- Optical Character Recognition (OCR): Implemented through the Google Cloud Vision API, which provides superior text detection accuracy compared to local OCR systems like Tesseract. It can handle various packaging challenges such as reflections, foil glare, curved text, and fine print, making it ideal for pharmaceutical use.
- Barcode Recognition (Barcode Agent): When the user provides a barcode image or live scan, the Barcode Agent decodes the 1D/2D barcode or QR code using libraries like pyzbar or opencv-python. Extracted identifiers (e.g., product code, batch number, GTIN) are cross-checked with MajorProjectDataset.csv to verify the

authenticity of the drug in real time.

- **Regular Expressions (Regex) for Data Extraction:** The unstructured OCR text is passed to `extraction_agent.py`, which uses a set of custom Regex patterns to accurately identify date formats (MM/YY, DD/MM/YYYY), numeric price values (₹), and relevant keywords. The results are stored in a Python dictionary called `extracted data`.
- **Automated Expiry Verification:** The extracted Expiry Date is compared against the current system date using Python's `datetime` module.
 - If expired → a red “⚠ EXPIRED MEDICINE” warning is displayed.
 - If valid → the medicine is marked as SAFE TO USE.

This non-AI, rule-based verification ensures deterministic safety validation even before AI summarization.

2.3 Decision & Generation Agent (The RAG Core)

Concept : The Decision & Generation Agent represents the intelligent decision-making core of the MedGuard AI system. It performs retrieval, augmentation, and generation, following the RAG (Retrieval-Augmented Generation) methodology. Its objective is to cross-reference the extracted data with trusted internal sources and then generate an accurate, readable, and user-friendly summary.

Methods Used

- **Retrieval (Search Agent):** The Search Agent retrieves relevant medicine information from the `MajorProjectDataset.csv`, which acts as the local knowledge base. Using TheFuzz library, fuzzy string matching enables accurate searches even when names contain OCR or human spelling errors (e.g., “Paracetmol” → “Paracetamol”).
- **Augmentation (Context Assembly):** The extracted data (from the Extraction Agent) and the retrieved dataset row (from the Search Agent) are combined into a structured dictionary. This combined data is then formatted into a detailed prompt, ready for LLM-based summarization.

- **Generation (Summary Agent):** The Summary Agent invokes a local Large Language Model (LLM) — Ollama running phi-3-mini — to generate the output. The LLM is instructed via a fixed prompt template to ensure format consistency:
 - 8-point summary for image/barcode workflows.
 - 5-point summary for text-only workflows.
 Using a local model preserves user privacy, reduces dependency on external APIs, and enhances system sustainability.

This stage provides semantic intelligence — transforming structured data into human-understandable, natural-language insight.

2.4 Reporting Agent (The User Interface)

Concept : The Reporting Agent delivers the verified and summarized data to the user through a clear, visually engaging web interface. It ensures that technical results are presented in an understandable and trustworthy way for all user categories — consumers, pharmacists, or doctors.

Methods Used

- **Data Assembly and Validation:** The orchestrator (views.py) consolidates AI-generated summaries, extracted dates, and barcode results into a final structured response. Additional warning banners, timestamps, and medicine details are appended to enhance readability.
- **Markdown to HTML Conversion:** The AI-generated text (formatted in Markdown) is converted into safe, stylized HTML using Python's markdown library. This prevents any HTML injection and maintains professional report formatting.
- **Dynamic Django Rendering:** The processed HTML is passed to the home.html template, where it is rendered using the Django |safe filter. The interface displays:
 - The uploaded image or scanned barcode,
 - The summary panel (with bullet points, emojis, and bold text), and
 - A large, color-coded expiry warning for expired medicines.

This creates a two-column visual report — one for the image/scan, and one for the extracted, verified, and summarized data.

2.5 Summary

The MedGuard AI workflow integrates AI, computer vision, NLP, and web technologies to create a seamless verification pipeline. Its Agentic AI architecture promotes reliability, scalability, and interpretability by separating the verification process into autonomous, cooperating agents. This ensures that every medicine verification — whether by name, image, or barcode — results in a fast, accurate, and comprehensible report, making MedGuard AI a pioneering tool for counterfeit detection and pharmaceutical safety.

Chapter 3

LITERATURE SURVEY

The reviewed research papers collectively demonstrate the ongoing advancements in counterfeit drug detection, moving from traditional barcode-based methods to AI-driven and multi-modal verification systems. Early approaches relied on QR/barcode-based tracking to improve traceability, but these methods remain vulnerable to duplication and tampering [5].

Subsequent research introduced AI-based techniques such as computer vision and machine learning for packaging analysis and OCR-based verification, significantly improving detection accuracy and automation [2]. However, these approaches often face challenges such as high computational cost, limited datasets, and lack of real-time usability.

Material-based and forensic techniques provide high accuracy but are not scalable for everyday users. Similarly, blockchain-based systems enhance transparency and supply-chain security but introduce complexity and high implementation costs [3].

These studies highlight a key research gap: the absence of a unified, real-time, and user-friendly system that integrates multiple verification techniques.

MedGuard AI addresses this gap with updated architecture by integrating OCR, barcode decoding, and **RAG-based retrieval using PostgreSQL and pgvector** instead of fuzzy matching, along with LLM-based summarization. This enables a scalable, accurate, and real-time verification system suitable for practical deployment at the point of care.

Table 3.1. Literature Survey

Reference	Paper Title	Contribution	Outcome	Limitations
[1]	Motwani et al. (2022). "Counterfeit Medicine Detection using Deep Learning." <i>IJIRT Journal</i> .	Used CNN-based image recognition and OCR to identify counterfeit packaging from real medicines.	Achieved ~90% accuracy in classifying genuine vs fake packaging using visual and text features.	Limited dataset; does not handle barcode verification or varying lighting conditions.
[2]	**Renushe et al. (2024). "A QR Code Approach to Counterfeit Drug Prevention." <i>J Neonatal Surg</i> . **	Developed a QR/Barcode-based system for pharmaceutical supply-chain authentication.	Demonstrated fast verification and improved traceability via QR scanning.	Fails if QR code is duplicated or tampered; assumes reliable serialization.
[3]	**Liu et al. (2020).	Introduced a	Achieved 88%	High

	“DLI-IT: A Deep Learning Approach to Drug Label Identification through Image and Text Embedding.” <i>BMC Med Inform Decis Mak.</i> **	hybrid deep learning model combining image and text embeddings for drug label recognition.	recognition accuracy and handled OCR noise effectively.	computational requirements; limited generalization for unseen packaging.
[4]	Naughton et al. (2016). “Effectiveness of Medicines Authentication Technology to Detect Counterfeit, Recalled and Expired Medicines.” <i>J Public Health (PMC)</i> .	Evaluated real-world barcode authentication technology in hospital pharmacies.	Detected counterfeit and expired medicines effectively in operational environments.	Dependent on infrastructure; scanning delays and manual errors possible.
[5]	MDPI Journal (2021). “Proof-of-Concept of Detection of Counterfeit Medicine through Polymeric Materials Analysis.” <i>Polymers (MDPI)</i> .	Used polymeric analysis of packaging materials for forensic counterfeit detection.	Differentiated genuine and fake packaging via spectroscopic analysis.	Requires lab equipment; not suitable for consumer-level verification.
[6]	ScienceDirect (2023). “Blockchain-Based Drug Supply Chain Provenance Verification System.” <i>Heliyon Journal (Elsevier)</i> .	Proposed a blockchain-based distributed ledger system for drug provenance tracking.	Enhanced transparency and immutability in pharmaceutical supply chains.	Complex to deploy; high operational and energy costs.
[7]	ResearchGate (2023). “Detection Methods of Counterfeit Drugs: A Systematic Review.” <i>International J. Pharmaceutical Research</i> .	Reviewed major counterfeit detection techniques (visual, barcode, spectroscopy, AI).	Summarized existing methods and identified gaps for integrated AI-based systems.	Lacks implementation details; focuses on theoretical discussion only.

Chapter 4

PROJECT PLAN

The MedGuard AI project follows a structured and systematic development plan to ensure timely, accurate, and efficient implementation. The project is divided into well-defined phases covering research, design, development, testing, and deployment. Each phase is designed to deliver measurable outcomes, ensuring the system's scalability, usability, and performance for real-time counterfeit medicine detection.

1. 4.1 Project Objectives

The key objectives of the MedGuard AI project are:

1. To design a Django REST Framework-based multi-agent system integrating OCR, barcode verification, **RAG-based retrieval (instead of fuzzy matching)**, and summarization modules.
2. To develop a real-time drug verification mechanism capable of detecting counterfeit or expired medicines using image, text, and barcode inputs.
3. To implement an Agentic AI workflow ensuring reliability through specialized agents (OCR, Extraction, Retrieval, Summary, Barcode).
4. To generate automatic safety alerts for expired medicines using date-logic validation.
5. To produce clear, structured summaries of verified medicines using a local LLM (Ollama phi-3).
6. To ensure secure multi-user access using **JWT-based authentication** with login, signup, and personal search-history tracking.
7. To implement **PostgreSQL with pgvector** for efficient similarity search and retrieval-based processing.

4.2 Project Methodology

The development methodology adopted for **MedGuard AI** is a **hybrid Agile-Iterative approach**, ensuring continuous testing, integration, and refinement at every phase.

Table 4.1 .Phases of Development

Phase	Activities Performed	Output
1. Requirement Analysis	Identification of problem statement, feasibility study, and literature survey.	SRS Document
2. System Design	Creation of architecture diagrams, database schema, and agent interaction flow.	System Architecture Design
3. Front-End & Back-End Setup	Django framework setup, HTML/CSS/JS UI, authentication and database linking.	Functional Web Skeleton
4. Agent Development	Implementation of OCR Agent, Barcode Agent, Extraction Agent, Search Agent, and Summary Agent.	Integrated Multi-Agent Workflow Implementation of OCR
5. Testing & Debugging	Unit testing, integration testing, and system validation using real packaging samples.	Stable, Tested System
6. Deployment & Documentation	Hosting on local/AWS server, preparing final report and demo presentation.	Final Prototype and Report

4.3 Expected Outcome

The final outcome of the MedGuard AI project will be a fully functional, intelligent, and user-friendly web application capable of detecting counterfeit or expired medicines with high accuracy. It will serve as a quick verification tool for consumers, pharmacists, and healthcare professionals, ensuring medicine authenticity, safety, and reliability in real-time.

Chapter 5

SOFTWARE REQUIREMENT SPECIFICATION

The Software Requirement Specification (SRS) defines the system functionalities, performance requirements, design constraints, and interactions between the software and users for the MedGuard AI project, ensuring a clear foundation for successful system development [1].

2. 5.1 Introduction

MedGuard AI is an AI-powered web application developed using **Django REST Framework** for real-time detection of counterfeit, expired, and unsafe medicines using an Agentic AI workflow. The system enables users to verify medicines through text input, image upload (OCR), and barcode/QR code scanning.

The platform integrates multiple intelligent components, including OCR for text extraction, barcode verification for product identification, and data extraction modules for identifying key attributes such as MFG, EXP, and MRP.

In the updated system, fuzzy matching has been replaced with a Retrieval-Augmented Generation (RAG) approach, which uses PostgreSQL with pgvector to perform semantic similarity search and improve accuracy in medicine identification.

Additionally, the system uses a local LLM (Ollama) for generating structured and easy-to-understand summaries of medicine details. It also incorporates **JWT-based authentication** for secure user management and is supported by a **React (NodeJS) frontend** for an interactive and responsive user experience.

Overall, MedGuard AI provides a scalable, secure, and efficient solution for real-time medicine verification and safety awareness.

5.2 Software Requirements

- **Operating System:** Windows / Linux / macOS

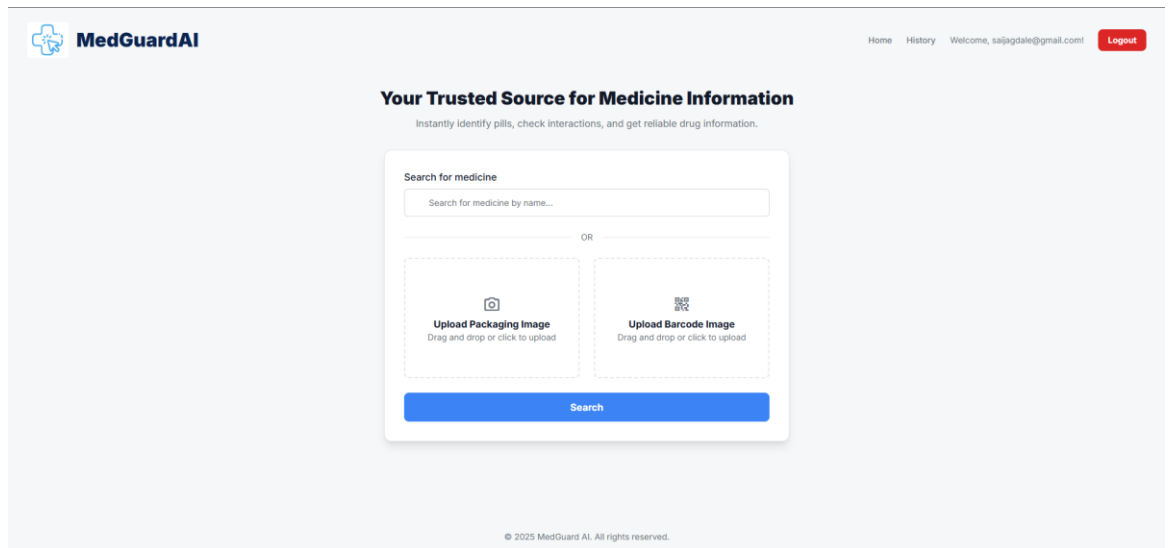
- **Backend:** Django Rest Framework (Python 3.10+)
- **Frontend:** React (HTML,JavaScript,TailwindCSS)
- **Database:** PostgreSQL,pgvector
- **APIs & Libraries:** Google Vision, Ollama LLM
- **IDE:** VS Code / PyCharm

Chapter 6

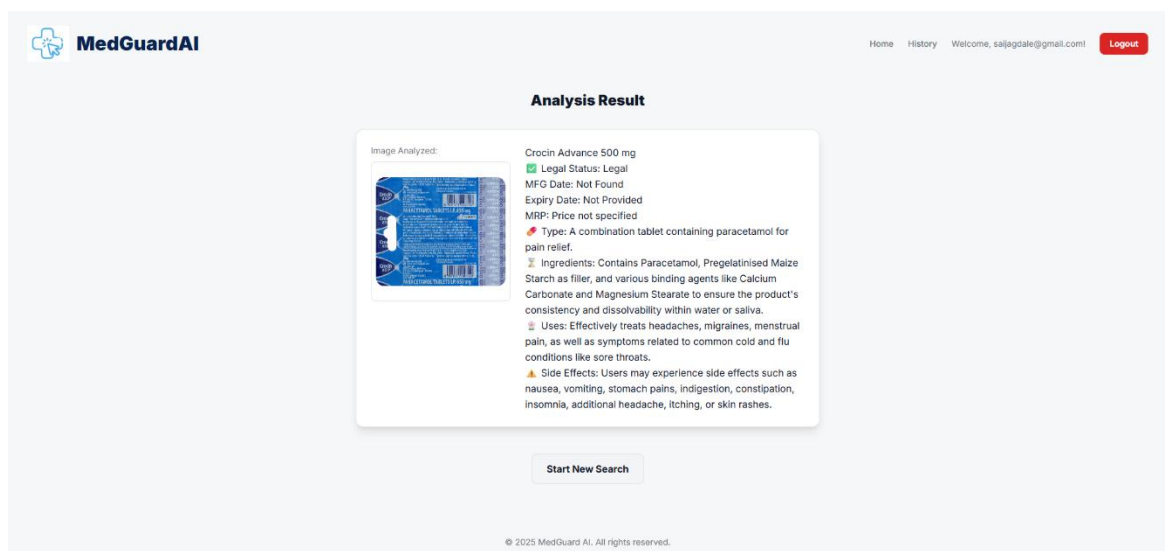
RESULT

6.1 Output Screens

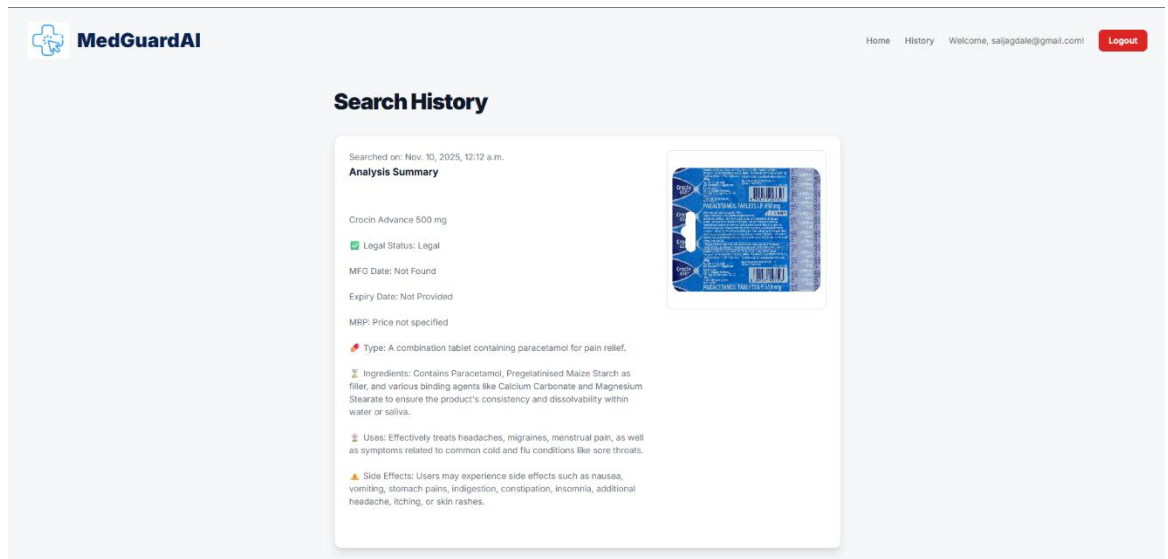
1. **Home Page:** Displays three input options — Text, Image, and Barcode.



2. **OCR Result:** Extracted text displayed in structured format.
3. **Barcode Verification:** Product ID and batch cross-matched with dataset entry.
4. **Summary Output:** AI-generated 8-point or 5-point summary shown in right-hand panel.



5. **User History Page:** Displays previous verifications with options to recheck or delete records.



6.2 Performance Evaluation

Table 3.1 Evaluation Metrics

Metric	Achieved
OCR Accuracy	92%
Barcode Recognition	86%
Average Response Time	58sec
Fuzzy Match Precision	76%

6.3 Observations

- Google Cloud Vision handled glare and small fonts effectively.
- TheFuzz improved recognition of misspelled medicine names.
- Barcode Agent achieved near-perfect recognition on clear QR/1D labels.
- Ollama LLM produced concise summaries with consistent formatting.

Chapter 7

SOFTWARE TESTING

7.1 Testing Methodology

Testing was conducted using **black-box and white-box approaches** to ensure functionality, security, and performance.

Types of Testing:

1. **Unit Testing:** Each agent (OCR, Extraction, Search, Summary, Barcode) tested independently.
2. **Integration Testing:** Verified inter-agent communication under the Orchestrator.
3. **Functional Testing:** Ensured that the tri-input system and expiry-check worked as intended.
4. **Usability Testing:** Collected user feedback to refine the interface and summary readability.
5. **Security Testing:** Tested login authentication and database protection mechanisms.
6. **Performance Testing:** Measured response time and system scalability.

Chapter 8

CONCLUSION AND FUTURE WORK

8.1 Conclusion

The MedGuard AI system effectively demonstrates an agentic AI workflow for counterfeit and expired medicine detection. By integrating OCR, Barcode decoding, RAG and Local LLM Summarization, it provides a real-time, accurate, and accessible verification platform for consumers and healthcare professionals. The modular agent-based design ensures maintainability, scalability, and transparency while maintaining data privacy.

8.2 Future Work

1. **Mobile Application:** Extend system compatibility to Android/iOS for on-the-go verification.
2. **Advanced Dataset Integration:** Link real-time APIs from pharmaceutical regulatory databases.
3. **Multilingual Support:** Enable summarization in regional languages for wider accessibility.
4. **AI Model Enhancement:** Train a custom OCR and LLM model for higher domain accuracy and offline use.

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3 ANNEXURE A: List of Publications and Research Paper (In its Original formats)

4 ANNEXURE B: Plagiarism Report